Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/09/2006
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
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Space Biology Cross-Element Discipline:	None		
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Comments:			
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: funding reinstated 8/2006 and will be a new grant, per D. Gillies (8/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Herlach, D. (Inst. fur Raumsimulation) Holland Mirtz, D. (Inst. fur Raumsimulation) Gangopadhyay, A. (Washington University) Hyers, R. (University of Massachussets)		
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Task Description:	Undercooling measurements and structural and physical property studies are made as a function of composition in Ti/Zr-TM liquids that form quasicrystals and related crystal phases. Quantitative measurements of the melt viscosity and specific heats are made as a function of alloy composition near the ideal stoichiometry of the icosahedrally-ordered phases in select Ti/Zr-alloys. These are needed to study the onset of icosahedral order in undercooled liquids and to accurately model nucleation as a function of alloy composition. They are particularly important for probing the role of oxygen in the stabilization of icosahedral order in the Ti-TM-Si-O alloys. The high liquidus temperatures and strong reactivities of the Ti/Zr-TM and Ti-TM-Si-O alloys prohibit accurate measurements of these quantities without containerless solidification. Ground-based investigations are, therefore, being made with the electrostatic levitator (ESL) located at Marshall Space Flight Center, to discern the relative roles of oxygen and diffusion on the nucleation process, make initial measurements of the physical properties, and determine the phase diagrams of the alloys studied. These data will allow more quantitative measurements of these physical properties and benchmark nucleation measurements to be made using the European Space Agency MSL-EML facility on the International Space Station. The ISS experiments will also allow the first investigations of the role of diffusion on nucleation in the absence of convection. The nucleation data will be compared with the predictions from a new model developed under partial support by this contract. This research focuses on studies of nucleation processes when diffusion effects are significant, on the formation and structures of highly ordered noncrystallographic phases, and on the connections between the atomic structures of these phases and the structures and thermophysical properties of liquids. The products of this research will lead to an improved understanding of nucleation and growth processes and will improve our ability to optimize desired microstructures for modern technological materials, both on earth and in reduced gravity environments. The new information gained will also provide valuable insight into the nucleation and growth mechanism of complex periodic and nonperiodic phases (i.e. quasicrystals), and how the liquid structure influences the microstructural development. Nucleation is the initial step in most first order phase transitions of interest to the physical and materials sciences. It is becoming increasingly clear that it plays a fundamental role in many transformations in biological and medical systems as well. An improved ability to control nucleation lies at the heart of the development of novel phase microstructures for advanced materials applications. The studies being carried out under the support of this NASA contract will expand our understanding of nucleation when the compositions of the initial and final phases are different, and when diffusion kinetics due to partitioning are competitive with interfacial attachment kinetics. They will lead to an improved understanding of the structures of undercooled liquids and polytetrahedral solid phases, of the growth of complex periodic and ordered nonperiodic phases, and of the mechanisms by which oxygen stabilizes icosahedral order in Ti-TM-Si-O alloys. The results of these studies will impact a number of technological areas in materials science, yielding an improved ability to develop new materials and phase microstructures and to better tune them to desired applications. The results of the ISS and ground-based research will be important for developing new materials for application to space exploration and materials processing in low gravity environments. In a low gravity environment, diffusion processes can dominate over the convective processes normally encountered with terrestrial processing of materials. It is essential, then, to develop a better understanding of microstructural development in such an environment; our studies of diffusion-influenced nucleation are a critical step toward that goal. As we have already demonstrated in our ground-based studies leading to the flight experiments, new techniques and technologies will also emerge that will impact broad areas of science and technology. Recently, working with representatives from NASA Marshall Space Flight Center, Huntsville, AL, and the Advanced Photon Source, we have developed a new technique (BESL) to obtain rapid structural information at high temperature on electrostatically levitated samples. This has already led to the verification of a half-century old hypothesis about nucleation from a metallic liquid. We have also demonstrated that such short-range order is common in metallic liquids and likely plays an important role in glass formation. This is important, since metallic glasses have been demonstrated to frequently have superior mechanical properties over crystalline materials. We have recently demonstrated that this technique can be applied to the rapid determination of the phase diagrams of high temperature materials, of central importance to the development of new materials for advanced propulsion for aerodynamic and space applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The BESL technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions.
	In March, 2005, our post-SCR ISS flight experiment, QUASI, was abruptly canceled, reflecting a change in NASA's mission. We have, therefore, spent the past year working on analyzing and publishing the ground-based results obtained in preparation for expected flight experiment. These include thermophysical property and nucleation data on supercooled liquids obtained with the electrostatic levitation (ESL) facility located at Marshall Space Flight Center and structural data obtained on the Advanced Photon Source (Argonne National Laboratory), using the novel facility (BESL, Beamline ESL) conceived and developed by us in the ground-based phase of QUASI. The key results obtained since the last report in March, 2005 are listed below. Aspects of some of the research activities compliment NSF sponsored research in Kelton's group at Washington University, supported by NSF DMR DMR 03-07410. The ESL and BSEL measurements discussed were made in collaboration with J. Rogers, R. Hyers and A. Goldman. The results obtained from these ground-based studies continue to have a significant scientific impact. 1. Previous measurements of the structure of supercooled liquid silicon by others had produced conflicting results, showing both increasing and decreasing coordination numbers with decreasing temperature. Using BESL we have recently obtained the most accurate measurements to date and to the lowest temperatures. These data clearly demonstrate that in conflict with previous studies there is no measurable change in the coordination number. Our results are also in conflict with previous theoretical treatments. Changes in the coordination number for liquid silicon upon supercooling, signaling an underlying liquid-liquid phase transition, have been widely predicted, based on molecular dynamics studies using a Stillinger-Weber potential. That the experimental data do not show the expected trend likely signals a subtle failure of this potential to accurately reflect the atomic interactions in liquid silicon. 2. Despite the constant coordination number, the structure factors, S(q), for undercooled liquid Si show subtle changes with increasing supercooling that indicate that the local environment is changing. We have recently developed a model of the local atomic structure based on a Reverse Monte Carlo analysis of the high-energy x-ray diffraction data (described in item 1) taken on a sample that was levitated using the electrostatic levitation technique. The resulting structures were investigated using Honeycutt and Andersen index and bond orientational order parameter analyses. While previous theoretical studies have suggested that the white-tin (A5) structure of the liquid evolves toward the

Task Progress:	diamond cubic (A4) with increased supercooling, we find that the amount of A5 and A4 order remains approximately fixed, but the coherence length of the A5 order increases. These results are consistent with the observed continuous growth of a double peak in the second oscillation of the pair correlation, the constant coordination number, and the constant nearest neighbor distance with supercooling. They indicate that the A5 structure is the preferred structure for the liquid, like the icosahedral structure is the preferred structure in the supercooled Ti-Zr-Ni liquids. 3. In earlier papers published under the support of this contract, we reported the first confirmation of a hypothesis made over a half-century ago, that the barrier to the crystallization of supercooled liquids arises from a developing icosahedral short range order in the liquid. A Ti-Zr-Ni alloy made near the composition of a eutectic that occurs at a Ni concentration of 21 at.%. was used for those studies. During the past year, we have published a manuscript in Physical Review B that presents a complete study of the maximum supercooling possible for a wide range of alloy compositions. We are preparing a manuscript for submission to Physical Review B that presents a discussion of the structures of the supercooled liquid in these alloys for the same range of composition. 4. It has been widely reported that glass formation improves in Zr62Cu20Ni8Al10 alloys when small amounts of Ti are substituted for Zr. Since glasses containing greater than 3 at.% Ti crystallize to an icosahedral quasicrystal phase, it has been argued that the Ti enhances icosahedral short range order in the liquid/glass. However, based on solidification and in-situ high-energy synchrotron diffraction studies of electrostatic-levitated supercooled liquids of these alloys, we have demonstrated that Ti inhibits surface crystallization, but neither increases the icosahedral short-range order nor improves glass formation. 5. We have made measured the density, viscosity, surface tension specific heats and thermal expansion of supercooled Ti-Zr-Ni liquids. These data show anomalous behavior in the supercooled liquid, above the recalescence temperature and any conceivable glass transition temperature. They suggest the presence of a novel liquid-liquid phase transition. We have also measured thermophysical properties for supercooled Zr54Ti8Cu20Al10Ni8 liquid alloys. Like the Ti-Zr-Ni, the surface tension shows a significant divergence from linearity as small undercoolings. 6. Working in collaboration with D. M. Herlach at the DLR in Cologne, Germany, we have made time-resolved in-situ studies of the phase formation sequence in supercooled Ti-Fe-Si-O liquids. Energy-dispersive x-ray measurements were made using synchrotron radiation on electromagnetically levitated liquid samples. Different solidification pathways were observed as a function of undercooling, with the primary nucleation of the 1/1 crystal approximant phase from the liquid occurring at the largest undercooling. That the nucleation of the 1/1 approximant phase is preferred over competing noncomplex phases such as a-Ti and b-Ti, leads to the conclusion that the approximant phase is the one with the lowest solid-liquid interfacial free energy. Similar to the results for the Ti-Zr-Ni alloys discussed in item (1) of this report, this suggests a growth of short-range icosahedral order with supercooling in Ti-Fe-Si-O.

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